



Connect a GitHub Repo with AWS



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```
6/8
Installed:
  git-2.47.1-1.amzn2023.0.2.x86_64      git-core-2.47.1-1.amzn2023.0.2.
2.47.1-1.amzn2023.0.2.noarch
  perl-Error-1:0.17029-5.amzn2023.0.2.noarch  perl-File-Find-1.37-477.amzn202
.1-1.amzn2023.0.2.noarch
  perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64  perl-lib-0.65-477.amzn2023.0.6.
Complete!
[ec2-user@ip-172-31-24-21 nextwork-web-project]$ git --version
git version 2.47.1
```



Introducing Today's Project!

In this project we will be connecting our web app project to a GitHub repo, making changes to the web app code and setting up a README file for our repo.

Key tools and concepts

I learnt the importance of pushing my my source code to my remote repository

Project reflection

This project took me approximately an hour and 30 minutes.

I would say the project really met my goals.

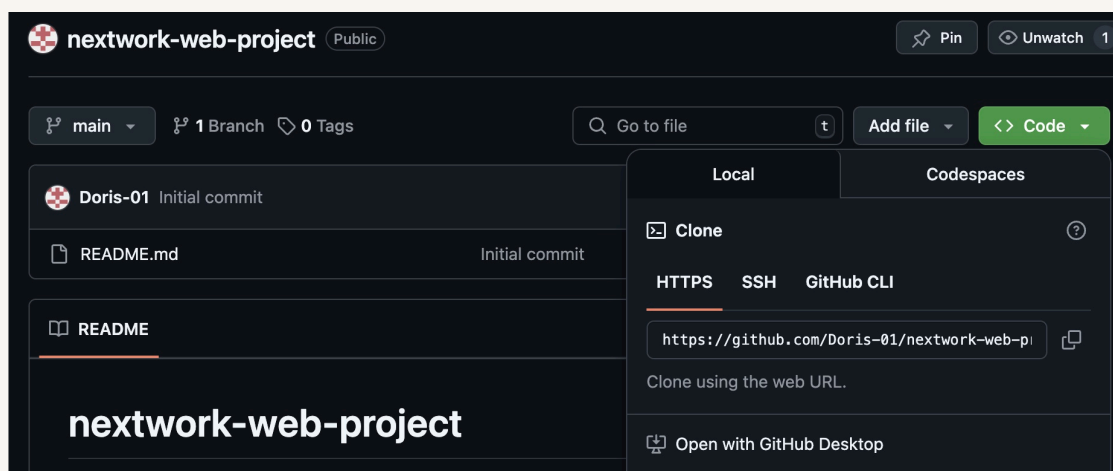
This project is part two of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project tomorrow.



Git and GitHub

Git is a version control system, It tracks every change you make, which lets you go back to an earlier version of your work if something breaks. I installed Git using the commands `sudo dnf update -y` `sudo dnf install git -y`

GitHub is a storage space for different version of your project that Git tracks. Since GitHub is a cloud service, it also lets you access your work from anywhere and collaborate with other developers over the internet.





My local repository

A Git repository is a folder that contains all your project files and their entire version history.

git init is a command that sets up the directory as a local Git repository which means changes are now tracked for version control. I ran git init in nextwork-web-project directory

After running git init, the response from the terminal was initialized empty git repository. A branch in Git is a parallel version of the same project.

```
ec2-user@ip-172-31-24-21 nextwork-web-project]$ ls
om.xml  src
ec2-user@ip-172-31-24-21 nextwork-web-project]$ git init
int: Using 'master' as the name for the initial branch. This default branch name
int: is subject to change. To configure the initial branch name to use in all
int: of your new repositories, which will suppress this warning, call:
int:
int:   git config --global init.defaultBranch <name>
int:
int: Names commonly chosen instead of 'master' are 'main', 'trunk' and
int: 'development'. The just-created branch can be renamed via this command:
int:
int:   git branch -m <name>
initialized empty Git repository in /home/ec2-user/nextwork-web-project/.git/
ec2-user@ip-172-31-24-21 nextwork-web-project]$
```




To push local changes to GitHub, I ran three commands

git add

The command I ran was remote add origin, telling Git where our GitHub repository is located. A staging area is where you're telling Git to put together all your modified files for a final review before you commit them.

git commit

git commit -m -saves the staged changes as a snapshot in your project's history. This means your project's version control history has just saved your latest changes in a new version. -m flag lets you leave a message

git push

git push -u origin master uploads your committed changes to origin, which you've bookmarked as your GitHub repo. By using -u you're also setting an 'upstream' for your local branch, which means you're telling Git to remember to push to master branch.



Authentication

When I commit changes to GitHub, Git asks for my credentials because Git needs to double check that I have the right to push any changes to the remote origin my local repo is connected with.

Local Git identity

Git needs my name and email because it needs author information for commits to track who made what change

We run git log to see your history of commits, which also mentions the commit author's name

```
ec2-user@ip-172-31-24-21 nextwork-web-project$ git log
commit f8f1aef36e2384e83efe64afe4869083b650f51f (HEAD -> master, origin/master)
Author: EC2 Default User <ec2-user@ip-172-31-24-21.ec2.internal>
Date: Mon Apr 14 04:33:23 2025 +0000

    Updated index.jsp with new content
```



GitHub tokens

GitHub authentication passed when I entered my personal access token

A GitHub token is a unique string of characters that looks like a random password. tokens are great for security because they would be difficult to guess.

I could set up a GitHub token by going to developer settings and creating a fine-grained classic token.





Making changes again

I wanted to see Git working in action, so I changed my index.jsp file. I couldn't see the changes in my GitHub repo initially because I hadn't pushed to my remote repo.

I finally saw the changes in my GitHub repo after I committed and pushed my changes.

```
<body>

  <h2>Hello Doris!</h2>

  <p>If you see this line in Github, that means your latest changes are getting pushed to your cloud repo

</body>
```



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